

Md. Taifur Hossain.

SOFTWARE ENGINEERING & ML / COMPUTER VISION INTERN

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github.com/Daceyreal · daceyreal.github.io

Final-year Computer Science undergraduate who ships working software. Built and released **four open-source tools** across 3D vision, model compression, and developer tooling — each with tests, CI, documentation, and a tagged release — plus a **first-author paper accepted at ICCA 2026 (ACM)**. Comfortable owning a problem end to end, from prototype to a documented release others can run. Seeking a software engineering or ML / computer vision internship.

TECHNICAL SKILLS

LANGUAGES	Python, C++, TypeScript / JavaScript, Bash
ML & VISION	PyTorch, NumPy, 3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, COLMAP, model quantization, RAG & embeddings (sentence-transformers)
WEB & UI	Three.js, Vite, Streamlit, HTML / CSS
TOOLS & INFRA	Git, Docker, GitHub Actions (CI/CD), GHCR, Linux, pytest, ruff

PROJECTS

splat-slim Python · Typer · NumPy · pytest + ruff CI

Command-line tool and library that compresses trained 3D Gaussian Splatting models by **~74% with under 3 dB quality loss**, with no retraining. Implemented a self-decodable mixed-precision quantization format that stores per-group ranges in a sidecar `.meta.npz`, so compressed files still load in standard viewers.

Shipped with automated tests and linting on GitHub Actions, MIT-licensed, with a tagged `v0.1.0` release.

nerf-prep Python · Docker · GitHub Actions · GHCR

Packaged a fragile multi-tool pipeline (COLMAP to Nerfstudio) into **one reproducible Docker command**, published to GHCR by CI. Handles matcher selection by capture type, GPU/CPU auto-detection with CPU fallback, headless operation, input validation, and a registration-rate summary.

Verified end to end on a real 40-photo capture: **40 of 40 images registered**.

splat-lab TypeScript · Three.js · Vite · live demo

Browser-based, real-time demo of the compression pipeline: load a scene, move sliders, and watch file size and visual quality trade off live on the GPU. Ported the Python compression math to TypeScript; runs entirely client-side with no backend.

paper-chat Python · Streamlit · sentence-transformers · PyMuPDF

Retrieval-augmented question answering over a PDF library, where **every answer cites the exact paper and page** it came from and declines when the sources do not cover the question. Swappable LLM backend (Groq or Gemini) behind a small interface, so the retrieval and citation logic is unit-tested with fakes and needs no API key to run.

PUBLICATION

Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

First author. Built the full experimental pipeline and evaluated it across three scenes on a single 16 GB GPU, reaching a 74% size reduction without retraining and diagnosing a quantization failure mode that a sub-group range scheme recovers to within 0.1–0.7 dB.

EDUCATION

International University of Business Agriculture & Technology (IUBAT) Dhaka, BD

B.Sc. in Computer Science & Engineering (BCSE)

Expected 2027

CGPA 3.4 / 4.00